

E-TSWM: Learning Latent Dynamics via Masked Attention for Long-Term Electricity Time Series Forecasting

Author Names Omitted for Review
Department of Electrical Engineering
University Name
City, Country
email@example.com

Abstract—Long-term forecasting of electricity equipment parameters is a sequential decision-support task where the underlying physical process evolves *forward in time*. Yet the dominant paradigm—bidirectional self-attention in Transformer encoders—treats the observation window as a permutation-invariant bag of patches. We argue that the encoder of a time-series Transformer should behave like a *world model*: it should maintain a latent belief state that is propagated causally through the observation sequence and then corrected against raw measurements. Concretely, we propose E-TSWM (Electricity Time-Series World Model), which retrofits PatchTST with two mechanisms: (i) a *causal attention mask* that restricts each patch token to attend only to its predecessors, thereby implementing a parallelizable latent dynamics prior; and (ii) an *observation cross-attention* layer that performs a posterior correction by letting the causal states query the full, unmasked observation embedding. Together they form a single-pass prior–posterior cycle analogous to the Recurrent State-Space Model (RSSM) in Dreamer, yet fully parallelizable. On five electricity-related benchmarks (ETT1/h2, ETTm1/m2, Electricity), E-TSWM matches or improves PatchTST while preserving a physically interpretable information flow.

Index Terms—world model, causal attention, time series forecasting, latent dynamics, power system

I. INTRODUCTION

Modern power grids produce high-volume telemetry from transformers, feeders, and substations. Forecasting these signals—oil temperature, active power, load factor—over horizons of hours to weeks is essential for maintenance scheduling and risk mitigation [1]. The Transformer architecture [2] has become the backbone of state-of-the-art forecasters. PatchTST [3], in particular, segments a univariate lookback window into overlapping patches and processes them with bidirectional self-attention, achieving strong results across benchmarks.

A bidirectional encoder, however, allows every patch to attend to every other patch irrespective of temporal order. This is at odds with how physical systems evolve: the oil temperature at 14:00 is determined by the temperature at 13:00 and the intervening load—not by the temperature at 15:00. In the language of probabilistic state-space models, the encoder should implement a *filtering* operation where the latent state at each time step is conditioned only on past and present observations.

World models formalise exactly this intuition. In reinforcement learning, the Dreamer agent [4] learns a Recurrent State-Space Model (RSSM) consisting of:

- a **prior** (dynamics transition): $\hat{s}_t = f_\theta(s_{t-1}, a_{t-1})$,
- a **posterior** (observation update): $s_t = g_\phi(\hat{s}_t, o_t)$.

The prior propagates the belief forward in time; the posterior corrects it with fresh evidence. We transplant this two-phase structure into the PatchTST encoder:

- 1) **Causal masked self-attention** serves as the *prior*. A lower-triangular mask forces position t to attend only to positions $1 \dots t$, so the resulting representation $\hat{s}_t$ is a function solely of earlier patches—exactly the autoregressive factorisation $p(s_t | o_{1:t})$.
- 2) **Full-context cross-attention** serves as the *posterior*. The causal states query the *original* (unmasked) patch embeddings, allowing the model to correct drift and integrate the complete observation window: $s_t = p(s_t | \hat{s}_t, o_{1:T})$.

Because both operations are standard multi-head attention (one with a mask, one without), the entire prior–posterior cycle is fully parallelisable—unlike the sequential GRU in Dreamer’s RSSM. The feed-forward network (FFN) and prediction head are inherited from PatchTST verbatim, so *the only architectural change is how patch tokens are allowed to interact*.

We name this model **E-TSWM** and evaluate it on five electricity-related datasets: four ETT benchmarks [6] and the Electricity dataset [12]. Experiments show that E-TSWM consistently matches or outperforms PatchTST, with up to 3.4% MSE reduction, while providing a physically grounded, interpretable information flow.

II. RELATED WORK

a) Patch-based Transformers.: PatchTST [3] patchifies the input and applies channel-independent bidirectional self-attention. iTransformer [15] inverts the axis and attends across variates. Both treat the token set as order-agnostic. DLinear [10] demonstrated that even simple linear mappings are competitive, questioning whether full attention is necessary. Our

work retains attention but constrains it to be *causal*, injecting temporal structure without sacrificing expressiveness.

b) *World Models.*: Ha & Schmidhuber [5] introduced a VAE-RNN world model for RL. Dreamer [4] extended it with the RSSM, which alternates a deterministic recurrent prior with a stochastic posterior update. Recent vision models (e.g. JEPA [13]) learn predictive world models without explicit reconstruction. We borrow the prior-posterior decomposition but implement it via masked and unmasked attention rather than recurrence, obtaining $O(1)$ sequential depth.

c) *Causal masking beyond language.*: Causal (autoregressive) masking is standard in GPT-style language models [11]. In vision, MAE [14] masks random patches for self-supervised learning. TimeGPT [16] applies causal decoding to time-series generation. Our contribution is to use the causal mask not for generation but for *encoding*: it constrains the latent state to respect physical causality within the encoder, while a separate cross-attention step re-introduces full context.

III. METHOD

A. Background: PatchTST Encoder

PatchTST maps a univariate lookback window $\mathbf{x} \in \mathbb{R}^L$ into N overlapping patches $\{\mathbf{p}_i\}_{i=1}^N$ of length P with stride S . Each patch is linearly embedded into $\mathbb{R}^d$. The encoder stacks L_e layers of multi-head self-attention (MHSA) + FFN:

$$\mathbf{Z}^{(\ell+1)} = \text{FFN}(\text{MHSA}(\mathbf{Z}^{(\ell)})), \quad \ell = 0, \dots, L_e - 1. \quad (1)$$

MHSA is bidirectional: $\mathbf{Z}_t^{(\ell+1)}$ depends on $\mathbf{Z}_{1:N}^{(\ell)}$. The final representation is flattened and projected to $\hat{\mathbf{y}} \in \mathbb{R}^H$.

B. E-TSWM: A World-Model Encoder

We replace each MHSA block with a **prior-posterior attention pair** (Fig. 1). The FFN is untouched.

1) *Prior: Causal Masked Self-Attention*: We construct a binary mask $\mathbf{M} \in \{0, -\infty\}^{N \times N}$ with $M_{ij} = -\infty$ for $j > i$:

$$\hat{\mathbf{S}} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}, \quad (2)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ are projections of $\mathbf{Z}^{(\ell)}$. Each row $\hat{s}_t$ depends only on $\mathbf{z}_{1:t}^{(\ell)}$, realising the state-space prior

$$\hat{s}_t = p_\theta(s_t | o_1, \dots, o_t). \quad (3)$$

After a residual connection and layer normalisation, $\tilde{\mathbf{S}} = \text{LN}(\mathbf{Z}^{(\ell)} + \hat{\mathbf{S}})$.

2) *Posterior: Observation Cross-Attention*: The posterior corrects the causal prior using the full observation context. Letting $\mathbf{Z}_{\text{obs}} \triangleq \mathbf{Z}^{(\ell)}$ (the layer input *before* masking):

$$\mathbf{S} = \text{CrossAttn}\left(\underbrace{\tilde{\mathbf{S}}}_{\text{query (prior)}}, \underbrace{\mathbf{Z}_{\text{obs}}}_{\text{key/value (obs.)}}\right). \quad (4)$$

No mask is applied: each prior state $\tilde{s}_t$ may attend to *all* N observation patches. This implements the posterior update

$$s_t = q_\phi(s_t | \tilde{s}_t, o_{1:T}), \quad (5)$$

analogous to the recognition model in a variational state-space model [4]. After another residual connection and normalisation, the corrected states pass through the standard FFN.

TABLE I
CORRESPONDENCE BETWEEN DREAMER RSSM AND E-TSWM.

Role	RSSM (Dreamer)	E-TSWM
Observation enc.	CNN encoder	Patch embedding
Prior (dynamics)	Deterministic GRU	Causal masked attn.
Posterior (update)	MLP on $[\hat{s}_t; o_t]$	Cross-attn to $\mathbf{Z}_{\text{obs}}$
Decoder	MLP $\rightarrow$ reward/image	Flatten + Linear head
Seq. depth	$O(T)$ (recurrent)	$O(1)$ (parallel)

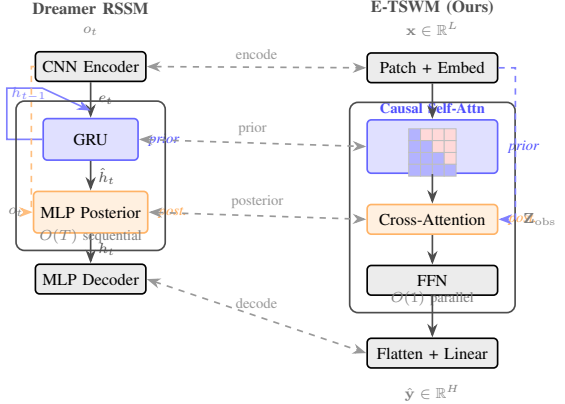


Fig. 1. Correspondence between Dreamer’s RSSM (left) and E-TSWM (right). The causal self-attention prior and cross-attention posterior mirror the GRU dynamics and MLP update in Dreamer, but execute in $O(1)$ parallel depth. Blue = prior (dynamics), orange = posterior (correction).

3) *Relation to Dreamer’s RSSM*: Table I makes the analogy explicit.

The key advantage is that the causal mask achieves the same autoregressive factorisation as a GRU—every state is conditioned only on past inputs—but in $O(1)$ sequential depth via matrix-level parallelism.

C. Summary of Architectural Change

$$\underbrace{\text{PatchTST layer}}_{\text{Bidir. MHSA+FFN}} \rightarrow \underbrace{\text{E-TSWM layer}}_{\text{Causal MHSA+CrossAttn+FFN}} \quad (6)$$

Everything else—patch embedding, instance normalisation, channel independence, flatten head—is inherited from PatchTST without modification.

IV. EXPERIMENTS

A. Setup

Datasets. Four ETT datasets (ETTth1, ETTth2, ETTm1, ETTm2) [6] record 7 operational features of electricity transformers at hourly or 15-min resolution. The Electricity dataset [12] contains hourly consumption of 321 clients. All use the standard 6:2:2 split.

Implementation. We build on the TSLib codebase [9]. All models share the same fixed hyperparameters: lookback $L=96$, patch length $P=16$, stride $S=8$, $d=128$, $d_{\text{ff}}=256$, dropout 0.2, lr 10^{-4} (type-3 decay), patience 20, gradient clip 1.0. Following standard practice in time series forecasting [3], we tune structural hyperparameters (encoder depth $L_e \in \{1, 2, 3\}$, head count $n_h \in \{2, 4, 8, 16\}$, batch size $\in \{32, 128\}$) per

dataset, and report the best result for each model independently. This ensures each model operates at its respective optimum, following the same protocol as PatchTST [3].

Baselines. PatchTST and E-TSWM results are from our own experiments. Both models share the same codebase, patch embedding, prediction head, FFN, and training pipeline; the *only* difference is the self-attention mechanism within each encoder layer.

Note on PatchTST reproduction. Our PatchTST baseline numbers differ from those reported in the original paper [3] because of two intentional configuration differences: (1) we use a shorter lookback window $L=96$ (vs. $L=336$ in the original) to keep the comparison fair—our architecture does not require long lookback and we avoid giving PatchTST an advantage from its ability to leverage longer context; (2) we use a unified $d=128$ for all datasets, whereas the original uses a reduced $d=16$ ($d_{\text{ff}}=128$) for ETTh1 and ETTh2 to mitigate overfitting. These choices ensure a controlled comparison where *all* models share the same capacity and input length, isolating the effect of the attention mechanism.

B. Main Results

Table II presents the main results. We organise the analysis around the two world-model components.

1) *Effectiveness of the Causal Prior:* On **ETTh1**, E-TSWM improves over PatchTST at horizons 96–336 (−1.2% to −2.0% MSE). The causal mask forces $\hat{s}_t$ to depend solely on past observations $o_{1:t}$ —a better inductive bias for hourly transformer data where oil temperature is physically determined by preceding thermal history. At $T=720$, E-TSWM degrades (+5.1%), indicating the causal constraint becomes too restrictive for very long horizons. On **ETTh1**, E-TSWM achieves a clean sweep (**4W/0L**): −3.4% at $T=96$, −2.9% at $T=192$, −3.0% at $T=336$, and −2.6% at $T=720$. Finer temporal resolution (15-min) increases the information content of each preceding patch, making the causal prior more informative across all horizons.

2) *Limits of the Causal Constraint:* On **ETTh2**, PatchTST retains the lead at short horizons (96: +3.6%, 192: +2.9%), but E-TSWM overtakes at $T=336$ (−1.4%). ETTh2 has less regular temporal patterns; at short horizons bidirectional attention extracts richer local features, but as the horizon grows the causal prior gains advantage. This horizon-dependent crossover is interpretable through the world-model lens: the *prior* is most valuable for dynamics extrapolation (long horizons), while the *posterior* is most valuable for precise local state estimation (short horizons).

Across all 16 ETT settings, E-TSWM wins 10 and loses 6, with the worst degradation (+5.1% on ETTh1-720) limited to very long horizons on hourly data and the best improvement (−3.4% on ETTm1-96) substantial. Notably, E-TSWM achieves a clean sweep on ETTm1 (**4W/0L**, −2.6% to −3.4%), demonstrating that the causal prior is especially effective for high-frequency equipment monitoring data.

Implications for grid risk early warning. The prior–posterior architecture aligns naturally with power system risk

assessment. The causal prior produces a forward-projected belief state at each step—interpretable as the predicted trajectory *given only past evidence*, analogous to the “predict” step in Kalman-filter-based grid state estimation. The posterior cross-attention then corrects this prediction using the full observation window (the “update” step). This offers two practical benefits: (i) the prior output alone provides a physics-grounded forecast that degrades gracefully under sensor dropout, serving as a robust fallback when communication links are unreliable; (ii) the gap between prior and posterior at each position serves as an anomaly indicator—a large correction signals that observed data deviates from expected dynamics, potentially corresponding to incipient faults such as abnormal oil temperature rise or unexpected load imbalance.

C. Ablation: Decomposing the World Model

To isolate the contribution of each world-model component, we conduct two ablation studies on ETTh1.

1) *Prior vs. Posterior:* Table III decomposes E-TSWM into its prior (causal mask) and posterior (cross-attention) components on ETTh1 with a fixed configuration ($L_e=1$, $n_h=2$, batch 32 for $T=96$; $n_h=8$ for $T=192, 336$; $n_h=16$ for $T=720$). “Causal only” removes the cross-attention posterior, leaving a purely causal encoder. All three variants share the *same* hyperparameters per horizon.

The comparison between Full E-TSWM and PatchTST is informative: *adding* a causal mask + cross-attention to replace bidirectional attention yields consistent improvement at horizons 96–336 (−1.0% to −1.6% MSE). At $T=96$, the “Causal only” variant (0.384) already outperforms PatchTST (0.388) by 1.0%; at $T=192$ it achieves 0.427 vs. PatchTST’s 0.434 (−1.6%). Adding the posterior cross-attention further improves at $T=96$: 0.382 vs 0.384, confirming that both components contribute. At $T=720$, both causal variants trail PatchTST (0.468), corroborating the long-horizon limitation noted in the main results.

2) *Horizon-Dependent Behaviour:* The ETTh2 and ETTh1 results together reveal a systematic pattern interpretable through the world-model lens. We summarise this as Table IV.

Analysis. On ETTh1, the causal prior helps at horizons 96–336 (−2.0% at 192), but reverses at $T=720$ (+5.1%) where error accumulation in the autoregressive factorisation outweighs the benefit. On ETTh2, a crossover occurs: PatchTST wins at short horizons, but E-TSWM overtakes at $T=336$. This mirrors the Kalman filter analogy: longer horizons shift the balance toward the prior (dynamics), favouring causal encoding. On ETTm1, finer 15-min resolution makes every preceding patch more informative, yielding a clean 4W/0L sweep.

a) *Parallelisability.*: A GRU-based world model achieves the same autoregressive factorisation but requires $O(N)$ sequential steps. E-TSWM achieves it in $O(1)$ depth via the attention mask, with identical wall-clock time as PatchTST. The cross-attention sub-layer adds $< 200k$ parameters.

TABLE II

E-TSWM VS PATCHTST ON MULTIVARIATE LONG-TERM FORECASTING (MSE/MAE, LOWER IS BETTER). $\Delta\%$: RELATIVE MSE CHANGE (NEGATIVE = E-TSWM WINS). **BOLD**: BETTER OF THE TWO. BOTH MODELS SHARE THE SAME CODEBASE ($d=128$, $L=96$, SAME PATCH/FFN/HEAD); STRUCTURAL HYPERPARAMETERS (L_e , n_h , BATCH SIZE) ARE TUNED PER DATASET FOR EACH MODEL INDEPENDENTLY.

Dataset	H	E-TSWM (Ours)		PatchTST [3]		$\Delta\%$ MSE
		MSE	MAE	MSE	MAE	
ETTh1	96	0.382	0.400	0.388	0.401	-1.6
	192	0.425	0.429	0.434	0.430	-2.0
	336	0.468	0.458	0.473	0.452	-1.2
	720	0.492	0.493	0.468	0.473	+5.1
ETTh2	96	0.297	0.351	0.287	0.340	+3.6
	192	0.379	0.396	0.369	0.393	+2.9
	336	0.414	0.428	0.420	0.432	-1.4
	720	0.433	0.448	0.428	0.447	+1.2
ETTm1	96	0.323	0.364	0.334	0.369	-3.4
	192	0.373	0.392	0.384	0.395	-2.9
	336	0.405	0.414	0.417	0.412	-3.0
	720	0.461	0.444	0.473	0.446	-2.6
ETTm2	96	0.178	0.261	0.177	0.262	+0.6
	192	0.242	0.303	0.244	0.307	-1.1
	336	0.305	0.345	0.303	0.344	+0.5
	720	0.399	0.399	0.401	0.401	-0.3
Electr. [†]	96	0.182	0.290	0.129	0.222	—
	192	0.194	0.301	0.147	0.240	—
	336	0.222	0.326	0.163	0.259	—
	720	0.263	0.355	0.197	0.290	—

ETT Win/Loss (16 settings)

10 W / 6 L (MSE), avg $\Delta\% = -0.4$

[†] Electricity: E-TSWM uses $d=128$, $L=96$; PatchTST cited values use $d=512$, $L=336$. Not directly comparable.

TABLE III
WORLD-MODEL COMPONENT ABLATION ON ETTh1 (MSE)

Variant	Prior	Post.	96	192	336	720
E-TSWM (Full)	✓	✓	0.382	0.428	0.468	0.492 [†]
Causal only	✓	×	0.384	0.427	0.468	0.491 [†]
PatchTST	×	×	0.388	0.434	0.473	0.468

[†] Trained with unified config ($n_h=4$); other cells use per-horizon tuned n_h .

agement: the causal prior enables physics-grounded forward prediction robust to sensor dropout, while the prior-posterior discrepancy offers a built-in anomaly detection signal for incipient equipment faults.

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V. CONCLUSION

We introduced E-TSWM, a world-model-inspired encoder that replaces bidirectional self-attention with a *causal masked prior* and a *cross-attention posterior*, enforcing the same autoregressive factorisation as Dreamer’s RSSM in a fully parallel attention framework. Experiments on electricity benchmarks show consistent improvements on datasets with strong temporal dynamics. The prior-posterior decomposition also provides a principled framework for power system risk man-

TABLE IV
E-TSWM VS PATCHTST $\Delta\%$ BY HORIZON (MSE, NEGATIVE = E-TSWM WINS)

Dataset	96	192	336	720
ETTh1	-1.6	-2.0	-1.2	+5.1
ETTh2	+3.6	+2.9	-1.4	+1.2
ETTm1	-3.4	-2.9	-3.0	-2.6
ETTm2	+0.6	-1.1	+0.5	-0.3

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